

84 Results

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Arts, Sciences, and Engineering, Spring 2025 : CS

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Enrollment Status: ● open ⊘ closed ⚠ waitlist

CS-0004 **Teaching Computer Science**

Mechanics of teaching, reviewing, and evaluating computer science coursework. Constructive question answering, grading with consistency, and creating an inclusive and welcoming environment. Enrolled students must be undergraduate teaching assistants for Computer Science classes.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22575	Regular	Time Not Specified TBA Medford/Somerville	Megan Monroe	1	●	⦿

Meeting Dates 01/15/2025 - 04/28/2025

Career AS&E Undergrad

Textbook [Textbook info](#)

Grading Pass/Not Pass

Add Consent Department Consent Required

Enrollment Reqs
None

Repeat for Credit No

Allow Multiple
Enroll in Term No

Total Units
Allowed 1

Total
Completions
Allowed 1

Instruction Mode In Person

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	25	10	15
Wait List	5	0	5

CS-0005 **Special Topics for non-majors - Preparing for Career Success**

Content and prerequisites to be announced. Please see department website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	25011	Regular	Tu 4:30PM - 6:30PM Eaton Hall, 201 Medford/Somerville	Karen Donoghue	2	●	⦿

Meeting Dates 01/15/2025 - 04/28/2025

Career AS&E Undergrad

Textbook [Textbook info](#)

Grading Graded

Enrollment Reqs
None

Section Class No. Session Day, Times and Locations Faculty Credit Status Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Notes This course is focused on helping Computer Science students prepare for finding roles in the technology industry. It covers topics such as strategic thinking about opportunities, skills assessment, marketing your skills, resume and marketing materials preparation, interview and practice, improving presentation skills, and career networking. Most class sessions will be virtual, with some class meetings in person. Students interested in registering for this class should fill out the form emailed to CS and DS students.							
Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		3					
Total Completions Allowed		1					
Instruction Mode		In Person					
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		30	29	1			

CS-0010 **Computer Science for All**

Fundamental concepts of computer science and computer programming for students with no prior programming experience but who are interested in taking CS11. Data structures and algorithms, text and data manipulation, simulation using random numbers. Students may only receive credit for CS10 if it is completed prior to CS11 and CS15. Students may receive credit for either CS10 or ES2, but not both. Recommendation: high school algebra.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22576	Regular	Mo, We 1:30PM - 2:45PM Braker Hall, Room 001 Medford/Somerville	Elyse L. Cornwall	4		
Meeting Dates		01/15/2025 - 04/28/2025					
Class Attributes		BFA-Science/Technology, SOE-Computing					
Career		AS&E Undergrad					
Textbook		Textbook info					
Grading		Graded					

Enrollment Reqs
First or second year student in AS&E or CS Postbac or Major or Minor in (Computer Science or Cognitive and Brain Science) in A&S.

Class Notes
Students who do not meet the enrollment requirements can sign up through the special waitlist section (02-LEC) and will be allowed into the class if space permits.

Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		4					
Total Completions Allowed		1					
Instruction Mode		In Person					

Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		125	105	20			
Wait List		10	0	10			

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
02-LEC Details	24562	Regular	Time Not Specified TBA Medford/Somerville	Elyse L. Cornwall	4		☐

CS-0011 **Introduction To Computer Science**

The study of computer science centers on two complementary aspects of the discipline. First, computer science is fundamentally concerned with the problem-solving methodologies it derives from its foundational fields: the design principles of engineering, mathematical theory, and scientific empirical study. Second, these methodologies are applied in the complex context of a modern day computing system. In this course we will address both of these important aspects. As a means for developing your design skills, we will discuss the fundamental features of a high level, general purpose programming language — namely C++ — and learn how to use it as a tool for problem solving. We will also consider the performance of solutions, and how to apply both analytical and empirical assessment techniques. Finally, we will explore the Unix operating system as a context for problem solving.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22887	Regular	Tu 5:30PM - 7:00PM Online	Partha Biswas	4		☒

Meeting Dates	01/15/2025 - 04/28/2025
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing
Career	AS&E Undergrad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
Open to online CS or DS postbac, OR online CS or DS MS students

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	4
Total Completions Allowed	1
Instruction Mode	Online

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	30	11	19
Wait List	5	0	5

CS-0011 **Introduction To Computer Science**

The study of computer science centers on two complementary aspects of the discipline. First, computer science is fundamentally concerned with the problem-solving methodologies it derives from its foundational fields: the design principles of engineering, mathematical theory, and scientific empirical study. Second, these methodologies are applied in the complex context of a modern day computing system. In this course we will address both of these important aspects. As a means for developing your design skills, we will discuss the fundamental features of a high level, general purpose programming language — namely C++ — and learn how to use it as a tool for problem solving. We will also consider the performance of solutions, and how to apply both analytical and empirical assessment techniques. Finally, we will explore the Unix operating system as a context for problem solving.

LECTURE

Select 1 Lecture, 1 Laboratory

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22577	Regular	Tu, Th 10:30AM - 11:45AM Joyce Cummings Center, 270 Medford/Somerville	Elyse L. Cornwall	4		☐

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
02-LEC Details	22578	Regular	Tu, Th 12:00PM - 1:15PM Joyce Cummings Center, 270 Medford/Somerville	Elyse L. Cornwall	4		☐
03-LEC Details	22579	Regular	Tu, Th 10:30AM - 11:45AM No Room Assigned Medford/Somerville	Elyse L. Cornwall	4		☐
04-LEC Details	22580	Regular	Tu, Th 12:00PM - 1:15PM No Room Assigned Medford/Somerville	Elyse L. Cornwall	4		☐

LABORATORY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
LA-LAB Details	22581	Regular	We 9:00AM - 10:15AM Joyce Cummings Center, 235 Medford/Somerville	Elyse L. Cornwall	0		☐
LC-LAB Details	22583	Regular	We 10:30AM - 11:45AM Joyce Cummings Center, 235 Medford/Somerville	Elyse L. Cornwall	0		☐
LD-LAB Details	22584	Regular	We 10:30AM - 11:45AM Joyce Cummings Center, 240 Medford/Somerville	Elyse L. Cornwall	0		☐
LE-LAB Details	22585	Regular	We 1:30PM - 2:45PM Joyce Cummings Center, 235 Medford/Somerville	Elyse L. Cornwall	0		☐
LF-LAB Details	22586	Regular	We 1:30PM - 2:45PM Joyce Cummings Center, 240 Medford/Somerville	Elyse L. Cornwall	0		☐
LG-LAB Details	22587	Regular	We 3:00PM - 4:15PM Joyce Cummings Center, 235 Medford/Somerville	Elyse L. Cornwall	0		☐
LI-LAB Details	22589	Regular	We 4:30PM - 5:45PM Joyce Cummings Center, 235 Medford/Somerville	Elyse L. Cornwall	0		☐
LW-LAB Details	22593	Regular	Time Not Specified TBA Medford/Somerville	Elyse L. Cornwall	0		☐

CS-0014 **Emerging Scholars in Computer Science**

Weekly, peer-led workshops exploring topics in computer science. Emphasis on the collaborative and problem-solving nature of computer science. No prior programming experience is necessary. Students must apply to enroll – see departmental website for details. Prerequisite: first year or sophomore standing

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	21751	OPEN	Time Not Specified TBA Medford/Somerville	Richard Townsend	1		☒

CS-0015 **Data Structures**

A second course in computer science. Data structures and algorithms are studied through major programming projects. Topics include linked lists, trees, graphs, dynamic storage allocation, and recursion. Enrollment priority given to freshmen or sophomores; computer science majors or minors; or majors or minors that list CS15 as a requirement or elective.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	21751	OPEN	Time Not Specified	Richard Townsend	1		☒

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22888	Regular	Th 5:30PM - 7:00PM Online	Chris Magnano	4		☒

CS-0015 Data Structures

A second course in computer science. Data structures and algorithms are studied through major programming projects. Topics include linked lists, trees, graphs, dynamic storage allocation, and recursion. Enrollment priority given to freshmen or sophomores; computer science majors or minors; or majors or minors that list CS15 as a requirement or elective.

LECTURE

Select 1 Lecture, 1 Laboratory

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22594	Regular	Mo, We 10:30AM - 11:45AM Barnum Dana Complex, Room LL08 Medford/Somerville	Milod Kazerounian	4		☐
02-LEC Details	22595	Regular	Mo, We 3:00PM - 4:15PM Robinson Wing TTC, Room 253 Medford/Somerville	Milod Kazerounian	4		☐
03-LEC Details	22596	Regular	Mo, We 10:30AM - 11:45AM No Room Assigned Medford/Somerville	Milod Kazerounian	4		☐
04-LEC Details	22597	Regular	Mo, We 3:00PM - 4:15PM No Room Assigned Medford/Somerville	Milod Kazerounian	4		☐

LABORATORY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
LA-LAB Details	22598	Regular	Tu 9:00AM - 10:15AM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LB-LAB Details	22599	Regular	Tu 9:00AM - 10:15AM Joyce Cummings Center, 240 Medford/Somerville	Milod Kazerounian	0		☐
LC-LAB Details	22600	Regular	Tu 10:30AM - 11:45AM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LD-LAB Details	22601	Regular	Tu 1:30PM - 2:45PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LE-LAB Details	22602	Regular	Tu 3:00PM - 4:15PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LF-LAB Details	22603	Regular	Tu 3:00PM - 4:15PM Joyce Cummings Center, 240 Medford/Somerville	Milod Kazerounian	0		☐
LG-LAB Details	22604	Regular	Tu 4:30PM - 5:45PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LH-LAB Details	22605	Regular	Tu 4:30PM - 5:45PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LI-LAB Details	22606	Regular	Tu 6:00PM - 7:15PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
LJ-LAB Details	22607	Regular	Tu 6:00PM - 7:15PM Joyce Cummings Center, 235 Medford/Somerville	Milod Kazerounian	0		☐
LW-LAB Details	22608	Regular	Time Not Specified TBA Medford/Somerville	Milod Kazerounian	0		☐

CS-0020 Web Programming

An introduction to techniques, principles, and practices of writing computer programs for the World Wide Web. Server and browser capabilities and limits. Media types, handlers, and limitations. Web programming languages and techniques. Web security, privacy, and commerce. Lectures augmented with programming projects illustrating concepts and current practice.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22609	Regular	Tu, Th 12:00PM - 1:15PM Joyce Cummings Center, 160 Medford/Somerville	Lisa Diorio	3		☒

CS-0021 Concurrent Programming

Shared memory concurrency including threads, locks, mutexes, semaphores, Deadlock, livelock, shared mutable state. Native concurrency including communicating processes, message send/receive, synchronous and asynchronous communication, isolated mutable state. Team project using students' choice of paradigm and topic. Recommendations: Two programming courses past CS 15.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22780	Regular	Tu, Th 4:30PM - 5:45PM Joyce Cummings Center, 402 Medford/Somerville	Mark A Sheldon	3		☒

CS-0023 Game Design

Processes, procedures, methodologies, and principles of game design. Managing teamwork, project scope and user experience. Paper and digital prototyping, including the use of software for programming, art, and audio production. Recommendations: CS 15

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22610	Regular	Tu 6:00PM - 9:00PM Joyce Cummings Center, 170 Medford/Somerville	Jason Wiser	3		☒

CS-0027 How Systems Fail

Failure of computer systems within the larger context of complex systems, including the power grid and aviation. Failures of algorithms and protocols, engineering and implementation, systems and applications, people and culture. Attacks, attack recovery, security, privacy, and attribution. Case studies of failures and attacks, including distributed denial of service, Meltdown, Spectre, and spear-phishing attacks.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22611	Regular	Tu, Th 10:30AM - 11:45AM Joyce Cummings Center, 170 Medford/Somerville	Milod Kazerounian	3		☒

CS-0028 Cyber-Security and Cyber-Warfare

(Cross-listed as PS 164) An interdisciplinary analysis of cybersecurity and cyberwar in the United States and other countries. Policymaking and intelligence aspects of cybersecurity and political science and the technical constraints of

computer networks and software. Exploit methods including packet analysis, password cracking, social engineering, reconnaissance, and malware analysis. State-backed and non-state actors engaged in cyber-espionage, counterintelligence, deterrence, and offensive cyber-operations. Prerequisites: Political Science 61 or Computer Science 11.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22781	Regular	Tu, Th 1:30PM - 2:45PM Joyce Cummings Center, 265 Medford/Somerville	Jeffrey Taliaferro, Ming Yan Chow	3		☒

CS-0030 **Programming for Data Science - Programming for Data Science**

Fundamentals of programming for data-intensive science. Data structures and algorithms for data manipulation, cleaning, and preparation. Design of data manipulation programs. Coding standards and practices. Use and creation of software libraries. Techniques for improving program performance. Examples drawn from data preparation and transformation, statistical data analysis, machine learning, deep learning, and deep data science including recommendation systems and trend analysis.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22903	Regular	We 6:00PM - 7:30PM Online	Alva Couch	4		☒

CS-0040 **Machine Structure & Assembly-language Programming**

Structure of machine-level data and code, including memory, cache, registers, machine arithmetic, and bitwise operations. Encapsulating machine functionality through interfaces and abstract data types. Structure of assembly code, relocatable object code, and binary machine code, and the translations between them. Applications of machine-level operations and code translations in programming projects. High demand (see "course notes" for signup procedure).

LECTURE

Select 1 Lecture, 1 Laboratory

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22612	Regular	Tu, Th 1:30PM - 2:45PM Robinson Wing TTC, Room 253 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	5		☐
02-LEC Details	24508	Regular	Time Not Specified TBA Medford/Somerville	STAFF	5		☐

LABORATORY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
LA-LAB Details	22613	Regular	Fr 10:30AM - 11:45AM Joyce Cummings Center, 235 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	0		☐
LB-LAB Details	22614	Regular	Fr 10:30AM - 11:45AM Joyce Cummings Center, 240 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	0		☐
LC-LAB Details	22615	Regular	Fr 12:00PM - 1:15PM Joyce Cummings Center, 235 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	0		☐
LD-LAB Details	22616	Regular	Fr 12:00PM - 1:15PM Joyce Cummings Center, 240 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	0		☐
LE-LAB Details	22617	Regular	Fr 1:30PM - 2:45PM Joyce Cummings Center, 235 Medford/Somerville	Mark A Sheldon, Noah Mendelsohn	0		☐
LF-LAB Details	22618	Regular	Fr 1:30PM - 2:45PM	Mark A Sheldon, Noah Mendelsohn	0		☐

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
LW-LAB	24509	Regular	Joyce Cummings Center, 240 Medford/Somerville	STAFF	0		☐
Details			Time Not Specified TBA Medford/Somerville				

CS-0061 **Discrete Mathematics**

(Cross-listed as MATH 61). Sets, relations and functions, logic and methods of proof, combinatorics, graphs and digraphs. Recommendations: MATH 32 or COMP 11 or permission of instructor.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22777	Regular	Mo, We, Fr 9:30AM - 10:20AM Joyce Cummings Center, 170 Medford/Somerville	Marty Allen	3		☐
Details							
02-LEC	22619	Regular	Mo, We, Fr 10:30AM - 11:20AM Joyce Cummings Center, 170 Medford/Somerville	Marty Allen	3		☐
Details							
M1-LEC	22889	Regular	Tu 7:00PM - 8:30PM Online	Michael Allan Jahn	3		☐
Details							

CS-0062 **Discrete Math Skills Workshop**

Small group, active learning workshop that provides students additional learning resources related to CS 61.

WORKSHOP

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-WKS	25019	Regular	Fr 10:30AM - 11:30AM Bromfield-Pearson, Room 005 Medford/Somerville	Marty Allen	1		☐
Details							
02-WKS	25020	Regular	Fr 11:30AM - 12:30PM Joyce Cummings Center, 610 Medford/Somerville	Marty Allen	1		☐
Details							
04-WKS	25022	Regular	Fr 12:00PM - 1:00PM Halligan Hall, Room 169 Medford/Somerville	Marty Allen	1		☐
Details							
05-WKS	25023	Regular	Fr 1:00PM - 2:00PM Halligan Hall, Room 169 Medford/Somerville	Marty Allen	1		☐
Details							
06-WKS	25024	Regular	Fr 3:00PM - 4:00PM Joyce Cummings Center, 610 Medford/Somerville	Marty Allen	1		☐
Details							

CS-0093 **Directed Study**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-IND	22695	Regular	Time Not Specified TBA Medford/Somerville	Abani Patra, Alva Couch, Chris Magnano, Daniel Jared Votipka, Dave Lillethun, Deborah Sunter, Diane L Souvaine, Donna Slonim, Elaine Schaertl Short, Elyse L.	0-3		☐
Details							

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
				Cornwall, Fahad Rafique Dogar, Hari Sundar, Jeffrey Foster, Jivko Sinapov, Johannes Peter Albert De Ruiter, Johes Bater, Karen Edwards, Lenore J Cowen, Liping Liu, Mark A Sheldon, Marty Allen, Matthias Scheutz, Megan Monroe, Megumi Ando, Michael C. Hughes, Milod Kazerounian, Ming Yan Chow, Noah Mendelsohn, Raja Raman Sambasivan, Remco K Chang, Reuth Mirsky, Richard Townsend, Robert Jacob, Saeed Mehraban, Soha Hassoun, Susan Landau, Vladimir Podolskii			
02-IND Details	25288	Regular	Time Not Specified TBA Medford/Somerville	Richard Townsend	0-3		

CS-0098 **Senior Capstone Project II**

Implementation and testing of the project designed in CS 97. Implementation tools, strategies, and platforms. Testing and debugging methodologies. Maintenance and release management. Legal, ethical, and social impacts of computing. Students must have taken CS 97 the previous semester and be continuing the same project.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22620	Regular	Fr 12:00PM - 2:45PM Barnum Dana Complex, Room 104 Medford/Somerville	Dave Lillethun	3		

CS-0099 **Internship Computer Science**

Study of approved topics in Computer Science in concert with an internship in computing or a related field. Prerequisites: Permission of instructor.

INTERNSHIP

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-INT Details	22691	Regular	Time Not Specified TBA Medford/Somerville	Ming Yan Chow	1		

CS-0105 **Programming Languages**

Principles and application of computer programming languages. Emphasizes ideas and techniques most relevant to practitioners, but includes foundations crucial for intellectual rigor: abstract syntax, lambda calculus, type systems, dynamic semantics. Case studies, reinforced by programming exercises. Grounding sufficient to read professional literature. CS 80 and CS 105 cannot both be taken for credit.

LECTURE

Select 1 Lecture, 1 Recitation

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22621	Regular	Mo, We 3:00PM - 4:15PM Joyce Cummings Center, 270 Medford/Somerville	Richard Townsend	5		

RECITATION

REGISTRATION

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
RA-RCT Details	22622	Regular	Th 10:30AM - 11:45AM Bromfield-Pearson, Room 005 Medford/Somerville	Richard Townsend	0		☐
RB-RCT Details	22623	Regular	Th 12:00PM - 1:15PM Bromfield-Pearson, Room 005 Medford/Somerville	Richard Townsend	0		☐
RC-RCT Details	22624	Regular	Th 1:30PM - 2:45PM Bromfield-Pearson, Room 005 Medford/Somerville	Richard Townsend	0		☐
RD-RCT Details	22625	Regular	Th 3:00PM - 4:15PM Aidekman Arts Center, Room 013 Medford/Somerville	Richard Townsend	0		☐
RE-RCT Details	22626	Regular	Th 4:30PM - 5:45PM Aidekman Arts Center, Room 013 Medford/Somerville	Richard Townsend	0		☐
RF-RCT Details	22713	Regular	Th 6:00PM - 7:15PM Joyce Cummings Center, 402 Medford/Somerville	Richard Townsend	0		☐
RG-RCT Details	22714	Regular	Th 7:30PM - 8:45PM Joyce Cummings Center, 402 Medford/Somerville	Richard Townsend	0		☐
RH-RCT Details	22627	Regular	Fr 10:30AM - 11:45AM Joyce Cummings Center, 402 Medford/Somerville	Richard Townsend	0		☐
RI-RCT Details	22628	Regular	Fr 12:00PM - 1:15PM Eaton Hall, 211 Medford/Somerville	Richard Townsend	0		☐
RJ-RCT Details	22715	Regular	Fr 1:30PM - 2:45PM Eaton Hall, 170 Medford/Somerville	Richard Townsend	0		☐

CS-0111 Operating Systems

(Cross-listed as EE 128). Fundamental issues in operating system design. Concurrent processes: synchronization, sharing, deadlock, scheduling. Relevant hardware properties of uniprocessor and multiprocessor computer systems.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22629	Regular	Tu, Th 4:30PM - 5:45PM Joyce Cummings Center, 170 Medford/Somerville	Dave Lillethun	3		☐
M1-LEC Details	22902	Regular	Tu 7:00PM - 8:30PM Online	Partha Biswas	3		☐

CS-0114 Network Security

Vulnerabilities, attacks, and mitigations at all layers of the network stack. Public and private key cryptography, confidentiality and authentication protocols, botnets, firewalls, intrusion detection systems, and communication privacy and anonymity. Recommendations: CS 40.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22716	Regular	Tu, Th 12:00PM - 1:15PM Joyce Cummings Center, 180 Medford/Somerville	Daniel Jared Votipka	3		☒

CS-0115 Database Systems

Fundamental concepts of database systems, including conceptual design, relational and object-oriented data models, query languages (SQL, QBE), and implementation issues (indexing, transaction processing, concurrent control). The concepts and algorithms covered encompass many of those used in commercial and experimental database systems. Other topics include distributed databases and distributed query processing. Recommendations: CS 40

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22846	Regular	Mo 7:00PM - 8:30PM Joyce Cummings Center, 270 Medford/Somerville	Cody Doucette	3		☐
M1-LEC Details	22890	Regular	Tu 7:00PM - 8:30PM Online	Cody Doucette	3		☐

CS-0116 Introduction to Security

A holistic and broad perspective on cyber security. Attacking and defending networks, cryptography, vulnerabilities, reverse engineering, web security, static and dynamic analysis, malware, forensics. Principles illustrated through hands-on labs and projects, including Capture The Flag (CTF) games.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22630	Regular	Tu, Th 4:30PM - 5:45PM Joyce Cummings Center, 270 Medford/Somerville	Ming Yan Chow	3		☐
M1-LEC Details	22891	Regular	We 5:30PM - 7:00PM Online	Ming Yan Chow	3		☐

CS-0119 Big Data

Collecting, processing, analyzing and acting on data at Internet scale with speed and complexity. Techniques and infrastructures developed for big data including parallel and distributed database systems, map-reduce infrastructures, scalable platforms for complex data types, stream processing systems, and cloud-based computing. Course blends theory, algorithms and practical (hands on) work.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22631	Regular	Mo, We 10:30AM - 11:45AM Joyce Cummings Center, 180 Medford/Somerville	Jitendra Singh	4		☐
M1-LEC Details	22900	Regular	Mo 5:30PM - 7:00PM Online	Jitendra Singh	4		☐

CS-0120 Web Programming and Engineering

Web applications as complex systems that deliver functionality to a large number of users, and exhibit unique behaviors and demands in terms of performance, scalability, usability, and security. How the web works, limitations of client-side and server-side technologies including frameworks and APIs, content optimization, and data persistence and storage. Projects will involve search, using the cloud infrastructure, location-based services, mobile web development, and using tools to assess the security and privacy of web applications.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22897	Regular	Tu 5:30PM - 7:00PM Online	Lisa Diorio	3		☒

CS-0121 Software Engineering

Core principles and ideas that enable development of large-scale software systems, with a focus on programming. Abstraction, modularity, design patterns, specification, testing, verification, and debugging.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22896	Regular	We 7:30PM - 9:00PM Online	Vitaly Yurik	3		

CS-0122 Parallel Computing

(Cross-listed as EE 155) Programming modern parallel computer architectures, especially GPUs and multi-core CPUs. Rationale for modern multi-core CPUs. Challenges of multi-threaded programming. High-performance software taking advantage of hardware caches, cache coherency, memory systems and parallel computation. Recommendations: EE 25 or CS 40.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22826	Regular	Mo, We 1:30PM - 2:45PM Anderson Wing TTC, Room 212 Medford/Somerville	Joel Grodstein	3		

CS-0125 Numerical Analysis

(Cross-listed as MATH 125.) Analysis of algorithms involving computation with real numbers. Interpolation, methods for solving linear and nonlinear systems of equations, numerical integration, methods for ordinary differential equations. Recommendations: MATH 51 and programming ability in a language such as C, C++, Fortran, Pascal, or Matlab.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	24835	Regular	Mo, We 1:30PM - 2:45PM Joyce Cummings Center, 265 Medford/Somerville	Seulip Lee	4		

CS-0131 Artificial Intelligence

History, theory, and computational methods of artificial intelligence. Basic concepts include representation of knowledge and computational methods for reasoning. One or two application areas will be studied, to be selected from expert systems, robotics, computer vision, natural language understanding, and planning.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22632	Regular	Mo, We 6:00PM - 7:15PM Joyce Cummings Center, 160 Medford/Somerville	Fabrizio Santini	3		
M1-LEC Details	22898	Regular	Mo 7:30PM - 9:00PM Online	Fabrizio Santini	3		

CS-0132 Computer Vision

Introduction to low and intermediate levels of classic and modern Computer Vision. How to design algorithms that process visual scenes to automatically extract information. Fundamental principles and important applications of computer vision, including image formation, processing, detection and matching features, image segmentation, and multiple views. Basics of machine learning and deep learning for computer vision. Recommendations: CS 160 and Math 165

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	23536	Regular	Mo, We 6:00PM - 7:15PM Eaton Hall, 201 Medford/Somerville	Roy Shilkrot	3		

CS-0135 Introduction To Machine Learning And Data Mining

An overview of methods whereby computers can learn from data or experience and make decisions accordingly. Topics include supervised learning, unsupervised learning, reinforcement learning, and knowledge extraction from large databases with applications to science, engineering, and medicine. Recommendations: CS 160 is highly recommended.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22633	Regular	Tu, Th 12:00PM - 1:15PM Barnum Dana Complex, Room LL08 Medford/Somerville	Michael C. Hughes	3		☐
M1-LEC Details	22892	Regular	Fr 5:30PM - 7:00PM Online	Chris Magnano	3		☐

CS-0138 Reinforcement Learning

Agents that learn, plan, and act in complex, uncertain environments. Varying topics that may include practical successes and challenges of reinforcement learning; common software libraries and packages used to implement and test reinforcement learning algorithms, and advanced topics such as transfer learning and deep reinforcement learning. Recommendations: CS 131 and/or CS 135. Proficiency in one of C++, Java, or Python.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	23474	Regular	Mo, We 3:00PM - 4:15PM Joyce Cummings Center, 160 Medford/Somerville	Jivko Sinapov	3		☐
M1-LEC Details	22899	Regular	Th 7:00PM - 8:30PM TBA Online	Jivko Sinapov	3		☐

CS-0142 Network Science

Methods for analyzing massive graphs and networks that arise from social networks and biological networks. Scalable methods for classification, clustering, centrality, and inference on real-world example datasets. Specialized methods for social networks including link prediction, spread of influence, and emergent communities. Page rank in networks of web pages. Term project to identify a network science problem and test solutions on a real-world dataset. Recommendations: MATH 70 or 72 or CS 135 or 160.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	23476	Regular	Mo, We 4:30PM - 5:45PM Joyce Cummings Center, 280 Medford/Somerville	Lenore J Cowen	3		☒

CS-0144 Iterative Mthd Machine Learnng

(Cross-listed as EE 143) Design and analysis of modern machine learning methods with emphasis on convex and nonconvex problems, and centralized, federated, and distributed computational architectures. Topics include convergence, complexities, contractions, fixed point theorems, and perturbation techniques; gradient descent and stochastic gradient descent in addition to accelerated methods including Polyak and Nesterov momentum, minibatching, and variance reduction. State of the practice methods will be covered including Adagrad, Autogard, Adam, sgdm with applications in image classification and document clustering. Recommendations: MATH 70 and CS 11

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	24094	Regular	Tu 1:30PM - 4:00PM Anderson Wing TTC, Room 309 Medford/Somerville	Usman Khan	3		☒

CS-0146 Computer Architecture

(Cross-listed w/ EE 126) Modern computer architecture, starting from caches and memory systems, basic 5-stage pipelines and progressing to out-of-order superscalar processors, multicore processors, and heterogeneous

processors. Techniques to maximize single-thread performance within the constraints of memory technology, power consumption, and the inherent instruction-level parallelism of applications. Recommendations: an undergraduate computer organization class such as EE25 or CS40

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22850	Regular	Tu, Th 7:30PM - 8:45PM Halligan Hall, Room 169 Medford/Somerville	Alexander Hankin	3		☒
Details							

CS-0150 Special Topics - HCI for Disability

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22816	Regular	Tu, Th 9:00AM - 10:15AM Joyce Cummings Center, 280 Medford/Somerville	Elaine Schaertl Short	3		☐
Details							

CS-0150 Special Topics - Algorithmic Music Composition

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
02-LEC	23576	Regular	Mo, We 10:30AM - 11:45AM Joyce Cummings Center, 402 Medford/Somerville	Richard Townsend	3		☐
Details							

CS-0150 Special Topics - Generative AI for Social Impac

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
04-LEC	23662	Regular	Mo, We 10:30AM - 11:45AM 574 Boston Avenue, Room 401 Medford/Somerville	Fahad Rafique Dogar	3		Enrolled
Details							

CS-0150 Special Topics - Multi-Agent Systems

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
05-LEC	23745	Regular	Tu, Fr 12:00PM - 1:15PM Tisch Library, Room 316 Medford/Somerville	Reuth Mirsky	3		☐
Details							

CS-0150 Special Topics - MITRE eCTF

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
06-LEC Details	24992	Regular	Time Not Specified TBA Medford/Somerville	Ming Yan Chow, Steven Bell	3		☐

CS-0150 Special Topics - Entrepreneurship for Comp Sci

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
07-LEC Details	23488	Regular	Tu 5:15PM - 8:15PM Joyce Cummings Center,140 Medford/Somerville	Gavin Finn	3		☐

CS-0150 Special Topics - Topics in Computational Complexity

Content and prerequisites to be announced. Over the past three years, special topics courses have been offered in parallel computing, graph drawing, computational geometry, multimedia data compression, cryptography and security, digital network communication, spoken language systems, system and network administration, and machine learning. Please see departmental website for specific details.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
08-LEC Details	24543	Regular	Tu, Th 12:00PM - 1:15PM Joyce Cummings Center, 614 Medford/Somerville	Vladimir Podolskii	3		☒

CS-0151 Special Topics in Data Infrastructure and Systems - Addressing Cyber Threats, Vulnerabilities and Risk

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22817	Regular	Tu, Th 10:30AM - 11:45AM Anderson Wing TTC, Room 212 Medford/Somerville	Laurin Benedikt Weissinger	3		Enrolled

CS-0151 Special Topics in Data Infrastructure and Systems - Cybersecurity Clinic

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
02-LEC Details	22818	Regular	Mo 5:30PM - 7:00PM Joyce Cummings Center, 402 Medford/Somerville	Ming Yan Chow	3		☐

CS-0151 **Special Topics in Data Infrastructure and Systems - Privacy, Security, and Data**

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
03-LEC Details	23740	Regular	Tu, Th 4:30PM - 5:45PM Joyce Cummings Center, 280 Medford/Somerville	Johes Bater	3		In Cart

CS-0151 **Special Topics in Data Infrastructure and Systems - Anonymous Communications Theor**

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
04-LEC Details	23572	Regular	Tu, Th 1:30PM - 2:45PM Tisch Library, Room 310 Medford/Somerville	Megumi Ando	3		In Cart

CS-0151 **Special Topics in Data Infrastructure and Systems - Quantum Computer Science**

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
05-LEC Details	23573	Regular	Mo, We 1:30PM - 2:45PM Joyce Cummings Center, 280 Medford/Somerville	Saeed Mehraban	3		☐

CS-0151 **Special Topics in Data Infrastructure and Systems - Debugging Cloud Computing**

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
06-LEC Details	23574	Regular	Mo, We 4:30PM - 5:45PM Joyce Cummings Center,140 Medford/Somerville	Raja Raman Sambasivan	3		☐

CS-0151 **Special Topics in Data Infrastructure and Systems - Sustainable Computing**

A special topics course in data infrastructures and systems, suitable for fulfilling requirements of the Bachelor of Science in Data Science.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
07-LEC Details	24688	Regular	Tu, Th 9:00AM - 10:15AM Joyce Cummings Center, 160 Medford/Somerville	Hari Sundar	3		☒

CS-0160 **Algorithms**

Introduction to the study of algorithms. Strategies such as divide-and-conquer, greedy methods, and dynamic programming. Graph algorithms, sorting, searching, integer arithmetic, hashing, and NP-complete problems. High demand (see "course notes" for signup procedure).

LECTURE

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	22893	Regular	We 7:00PM - 9:00PM Online	Andrew Scott Winslow	4		☒

CS-0160 Algorithms

Introduction to the study of algorithms. Strategies such as divide-and-conquer, greedy methods, and dynamic programming. Graph algorithms, sorting, searching, integer arithmetic, hashing, and NP-complete problems. High demand (see "course notes" for signup procedure).

LECTURE

Select 1 Lecture, 1 Recitation

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22635	Regular	Mo, We 9:00AM - 10:15AM Barnum Dana Complex, Room LL08 Medford/Somerville	Karen Edwards	4		Enrolled
02-LEC Details	24513	Regular	Time Not Specified TBA Medford/Somerville	Karen Edwards	4		☐

RECITATION

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
RA-RCT Details	22636	Regular	Th 12:00PM - 1:15PM Aidekman Arts Center, Room 013 Medford/Somerville	Karen Edwards	0		☐
RB-RCT Details	22637	Regular	Th 12:00PM - 1:15PM Braker Hall, Room 113 Medford/Somerville	Karen Edwards	0		☐
RC-RCT Details	22638	Regular	Th 1:30PM - 2:45PM Anderson Wing TTC, Room 309 Medford/Somerville	Karen Edwards	0		☐
RD-RCT Details	22639	Regular	Th 1:30PM - 2:45PM Aidekman Arts Center, Room 013 Medford/Somerville	Karen Edwards	0		Enrolled
RE-RCT Details	22719	Regular	Th 3:00PM - 4:15PM Eaton Hall, 215 Medford/Somerville	Karen Edwards	0		☐
RF-RCT Details	22693	Regular	Th 3:00PM - 4:15PM Eaton Hall, 215 Medford/Somerville	Karen Edwards	0		☐
RG-RCT Details	22694	Regular	Th 4:30PM - 5:45PM Joyce Cummings Center, 610 Medford/Somerville	Karen Edwards	0		☐
RW-RCT Details	24514	Regular	Time Not Specified TBA Medford/Somerville	STAFF	0		☐

CS-0167 Computational Biology

(Cross listed as BME 167) Computational challenges in molecular biology, including sequence alignment and comparison, genomic annotation, transcriptomic data analysis, and proteomics. Underlying computational techniques such as dynamic programming, hidden Markov models, statistical analyses, and search and optimization procedures.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22640	Regular	Tu, Th 10:30AM - 11:45AM Joyce Cummings Center, 160 Medford/Somerville	Donna Slonim	3		☒

CS-0170 Computation Theory

CS-0170 Computation Theory

(Cross-listed as MATH 191). Models of computation: Turing machines, pushdown automata, and finite automata. Grammars and formal languages, including context-free languages and regular sets. Important problems, including the halting problem and language equivalence theorems.

LECTURE**Select 1 Lecture, 1 Recitation**

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22641	Regular	Tu, Th 1:30PM - 2:45PM Joyce Cummings Center, 270 Medford/Somerville	Megan Monroe	3		In Cart

RECITATION

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
RA-RCT Details	22642	Regular	We 9:00AM - 10:15AM Joyce Cummings Center, 180 Medford/Somerville	Megan Monroe	0		In Cart
RB-RCT Details	22643	Regular	We 12:00PM - 1:15PM Joyce Cummings Center, 302 Medford/Somerville	Megan Monroe	0		☐
RC-RCT Details	22644	Regular	We 3:00PM - 4:15PM Anderson Wing TTC, Room 312 Medford/Somerville	Megan Monroe	0		☐
RD-RCT Details	22645	Regular	We 4:30PM - 5:45PM Anderson Wing TTC, Room 312 Medford/Somerville	Megan Monroe	0		☐
RE-RCT Details	22646	Regular	We 6:00PM - 7:15PM Joyce Cummings Center, 170 Medford/Somerville	Megan Monroe	0		☐

CS-0171 Human Computer Interaction

Introduction to human-computer interaction, or how computers communicate with people. Methodology for designing and testing user interfaces, interaction styles (command line, menus, graphical user interfaces, virtual reality), interaction techniques (voice, gesture, eye movement), design guidelines, and user-interface management system software. Students will design a small user interface, program a prototype, and test the result for usability. Recommendations: CS 15 or junior/senior in CBS major

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22721	Regular	Mo, We 1:30PM - 2:45PM Joyce Cummings Center, 160 Medford/Somerville	Robert Jacob	3		☒

CS-0178 Visual Analytics

The science of combining interactive visual interfaces with data science, machine learning, and AI algorithms to support analytical reasoning. Principles, approaches, and tools for deriving insight from large, dynamic, ambiguous, and often conflicting data, and communicating findings effectively to aid in decision-making. Theoretical foundations illustrated via applying and developing visual analytics tools, as well as evaluating the effectiveness of these tools. Recommendations: CS 20 or 120.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	23362	Regular	Tu, Th 1:30PM - 2:45PM Joyce Cummings Center, 260 Medford/Somerville	Remco K Chang	3		☒

CS-0183 Privacy in the Digital Age

(Cross-listed with DHP P237) This module will provide an introduction to the threats to and protections for privacy in the digital age, examining public and private sector threats, and looking at issues from an international point of view. Topics to be covered include privacy threat models, location tracking and first and third party collection by private parties, government threats to privacy, and privacy protective technologies. No programming background needed, but a willingness and interest to play with digital tools is required.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22647	Regular	We 3:00PM - 5:20PM Cabot Intercultural Ctr, 205 Medford/Somerville	Thomas Cao	3		In Cart
Details							

CS-0191 **Research**

CS-0193 **Directed Study**

CS-0193 **Directed Study - Geometric Folding Algorithms**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
02-IND	24998	Regular	Time Not Specified TBA Medford/Somerville	Diane L Souvaine	3		☐
Details							

CS-0193 **Directed Study - MS CoreComp**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
03-IND	25126	Regular	Time Not Specified TBA Medford/Somerville	Karen Edwards	0		☐
Details							

CS-0193 **Directed Study - Directed Study for MS CoreComp**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
04-IND	25266	Regular	Time Not Specified TBA Medford/Somerville	Karen Edwards	0		☐
Details							

CS-0193 **Directed Study - Quantum PCP Conjecture**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
05-IND	25277	Regular	Time Not Specified TBA Medford/Somerville	Saeed Mehraban	3		☐
Details							

CS-0193 **Directed Study - HCI with Foundation Models**

Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
06-IND Details	25524	Regular	Time Not Specified TBA Medford/Somerville	Elaine Schaertl Short	0-4		

CS-0197 Honors Thesis B

Honors Thesis Computer Science.Please see departmental website for specific details.

THESIS

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-THS Details	22651	Regular	Time Not Specified TBA Medford/Somerville	Abani Patra, Alva Couch, Chris Magnano, Daniel Jared Votipka, Dave Lillethun, Diane L Souvaine, Donna Slonim, Elaine Schaertl Short, Elyse L. Cornwall, Fahad Rafique Dogar, Hari Sundar, Jeffrey Foster, Jivko Sinapov, Johannes Peter Albert De Ruiter, Johes Bater, Karen Edwards, Kathleen Fisher, Lenore J Cowen, Liping Liu, Mark A Sheldon, Marty Allen, Matthias Scheutz, Megan Monroe, Michael C. Hughes, Milod Kazerounian, Ming Yan Chow, Noah Mendelsohn, Raja Raman Sambasivan, Remco K Chang, Reuth Mirsky, Richard Townsend, Robert Jacob, Saeed Mehraban, Soha Hassoun, Susan Landau, Vasanth Sarathy, Vladimir Podolskii	3		

CS-0203 How Systems Fail

(Cross-listed w/ DHP D293) Graduate version of CS 27. Failure of computer systems within the larger context of complex systems, including the power grid and aviation. Failures of algorithms and protocols, engineering and implementation, systems and applications, people and culture. Attacks, attack recovery, security, privacy, and attribution. Case studies of failures and attacks, including distributed denial of service, Meltdown, Spectre, and spear-phishing attacks.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	22648	Regular	Tu, Th 10:30AM - 11:45AM Joyce Cummings Center, 170 Fr 10:30AM - 11:45AM Joyce Cummings Center, 302 Medford/Somerville	Milod Kazerounian	3		

CS-0239 Ethics for AI, Robotics, and Human-Robot Interaction

Technical challenges of endowing autonomous artificial agents with normative principles that will allow them to operate successfully in human societies. Algorithmic approaches in artificial agents (rule-based, utility-based, behavior-based, etc.) and their links to philosophical foundations of the main ethical theories (virtue ethics, deontology, utilitarianism). Conceptual and mathematical analysis of assumptions underlying each algorithmic approach as well as

implementation in agent-based simulations. Functional and performance tradeoffs of the algorithms and their implications for autonomous robots and AI systems. Mathematical and computational challenges of the different proposals for "implicit" and "explicit" ethical agents, including inverse reinforcement learning, verification-based approaches, and model checking.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22783	Regular	Mo, We 9:00AM - 10:15AM Anderson Wing TTC, Room 211 Medford/Somerville	Vasanth Sarathy	3		
Details							

CS-0260 **Advanced Algorithms**

(Cross-listed as MATH 221) Design and analysis of sequential, parallel, probabilistic, and approximation algorithms. Graph algorithms, sorting, searching, geometric algorithms, mathematical programming, lower bounds, and intractable problems. Recommendations: CS 160

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
1-LEC	23531	Regular	Mo, We 3:00PM - 4:15PM Joyce Cummings Center, 302 Medford/Somerville	Lenore J Cowen	3		
Details							

CS-0288 **Master of Science Capstone Project I**

Part one of a two-course, hands-on, and project-based culmination to the Master of Science in Computer Science Online program. Application of principles, strategies, methods, and tools for requirements analysis and design of a programming project, including project planning, project management, and proof of concept prototyping. Formulation of a project plan, including estimation of project completion requirements and timeline. To be taken in the second-to-last term of the Master of Science in Computer Science Online degree. Not available to students outside that program. Prerequisites: CS 180 or 121, and enrollment in the Master of Science in Computer Science Online program.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-IND	22895	Regular	Th 6:00PM - 7:00PM Online	Marty Allen	3		
Details							

CS-0289 **Master of Science Capstone Project II**

Part two of a two-course, hands-on, project-based culmination experience for the Master of Science in Computer Science Online program. Implementation of the project defined in part one, including use of principles, tools, and strategies for implementation, debugging, testing, documentation, maintenance, and release management. Presentation of final project results and documentation. To be taken in the last term of the Master of Science in Computer Science Online degree. Not available to students outside that program. Prerequisites: CS 288, and enrollment in the Master of Science in Computer Science Online program.

INDEPENDENT STUDY

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-IND	22649	Regular	Th 7:00PM - 8:00PM Online	Marty Allen	3		
Details							

CS-0291 **Seminar in Computer Science**

A weekly seminar with guest speakers discussing research challenges and recent advances in Computer Science. Pass/fail only. Satisfactory completion requires students attend and provide feedback for at least 50% of the seminars.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	22660	Regular	Th 3:00PM - 4:15PM Joyce Cummings Center, 270 Medford/Somerville	Hari Sundar, Johnny Redmond, Sandra Schulenburg	0		
Details							
02-LEC	22722	Regular	Fr 1:30PM - 2:30PM	Jivko Sinapov	0		
Details							

Section Details	Class No.	Session	Day, Times and Locations Robinson Wing TTC, Room 253 Medford/Somerville	Faculty	Credit	Status	Select
CS-0293	Master's Project						
CS-0296	Masters Thesis						
CS-0297	Dissertation Research						
CS-0299	Internship In Computer Science						
Study of approved topics in Computer Science in concert with an internship in computing or a related field outside the University. Prerequisites: Permission of instructor							
INTERNSHIP							
Section Details	Class No.	Session	Day, Times and Locations Time Not Specified TBA Medford/Somerville	Faculty Ming Yan Chow	Credit 1	Status 	Select
CS-0401	Masters Degree Continuation						
Part-time.Please see departmental website for specific details.							
CONTINUANCE							
Section Details	Class No.	Session	Day, Times and Locations Time Not Specified TBA Medford/Somerville	Faculty STAFF	Credit 0	Status 	Select
CS-0402	Masters Degree Continuation						
Full-time.Please see departmental website for specific details.							
CONTINUANCE							
Section Details	Class No.	Session	Day, Times and Locations Time Not Specified TBA Medford/Somerville	Faculty STAFF	Credit 0	Status 	Select
CS-0404	Graduate Engineering Co-op						
PRACTICUM							
Section Details	Class No.	Session	Day, Times and Locations Time Not Specified TBA Medford/Somerville	Faculty Alicia Abdulrazzaq	Credit 0	Status 	Select
CS-0405	Grad Teaching Assistant						
CS-0406	Grad Research Assistant						
CS-0501	Doctoral Degree Continuation						
Part-time.Please see departmental website for specific details.							
CONTINUANCE							
Section Details	Class No.	Session	Day, Times and Locations Time Not Specified TBA Medford/Somerville	Faculty STAFF	Credit 0	Status 	Select
CS-0502	Doctoral Degree Continuation						

Full-time.Please see departmental website for specific details.

CONTINUANCE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
FT-CON Details	22667	Regular	Time Not Specified TBA Medford/Somerville	STAFF	0		☒